

## Chapter 2

# Topological Spaces

In **Lecture 1** we saw that continuity between metric spaces can be characterized purely in terms of open sets (Theorem 3.4 of the Harris handout):  $f$  is continuous iff  $f^{-1}(\text{open}) = \text{open}$ . This means the metric itself is extraneous for the purposes of continuity, limits, and connectedness—all that matters is which sets are declared open. This motivates the following abstraction.

## 2.1 The Definition

### Definition 2.1.1: Topology

A topology on a set  $X$  is a collection  $\mathcal{T} \subseteq \mathcal{P}(X)$  of subsets of  $X$ , called the open sets, satisfying:

- (i)  $\emptyset \in \mathcal{T}$  and  $X \in \mathcal{T}$ .
- (ii) The union of any subcollection of elements of  $\mathcal{T}$  is in  $\mathcal{T}$ .
- (iii) The intersection of any finite subcollection of elements of  $\mathcal{T}$  is in  $\mathcal{T}$ .

A topological space is a pair  $(X, \mathcal{T})$ . When the topology is clear, we just write  $X$ . A subset  $U \subseteq X$  is open if  $U \in \mathcal{T}$ .

### Example 2.1.1 (The metric topology)

If  $(X, d)$  is a metric space, the collection  $\mathcal{T} = \{U \subseteq X : U \text{ is open in the metric sense}\}$  is a topology on  $X$  (proved in Lecture 1). This is the metric topology induced by  $d$ .

### Example 2.1.2 (The discrete and indiscrete topologies)

On any set  $X$ : the discrete topology is  $\mathcal{T} = \mathcal{P}(X)$  (every subset is open). This is the metric topology for the discrete metric  $d(x, y) = 1$  if  $x \neq y$ . The indiscrete (or trivial) topology is  $\mathcal{T} = \{\emptyset, X\}$ . In the indiscrete topology, the only open sets are the empty set and  $X$  itself.

### Example 2.1.3 (The finite complement topology)

Let  $X$  be any set. Define  $\mathcal{T}_f = \{U \subseteq X : X \setminus U \text{ is finite}\} \cup \{\emptyset\}$ . This is a topology:  $\emptyset$  and  $X$  are in  $\mathcal{T}_f$  (since  $X \setminus X = \emptyset$  is finite). If  $\{U_\alpha\}$  is a collection of elements of  $\mathcal{T}_f$ , then  $X \setminus \bigcup U_\alpha = \bigcap (X \setminus U_\alpha)$ , which is an intersection of finite sets, hence finite. If  $U_1, \dots, U_n \in \mathcal{T}_f$ , then  $X \setminus \bigcap U_i = \bigcup (X \setminus U_i)$ , a finite union of finite sets, hence finite.

**Example 2.1.4** (Topologies on a three-element set)

Let  $X = \{a, b, c\}$ . Examples of topologies include  $\{\emptyset, X\}$  (indiscrete),  $\{\emptyset, \{a\}, X\}$ ,  $\{\emptyset, \{a, b\}, \{c\}, X\}$ ,  $\{\emptyset, \{a\}, \{b\}, \{a, b\}, \{b, c\}, X\}$ , and  $\mathcal{P}(X)$  (discrete). The collection  $\{\emptyset, \{a\}, \{b\}, X\}$  is not a topology: it is missing  $\{a\} \cup \{b\} = \{a, b\}$ .

**Note:-**

Why bother with topologies that don't come from a metric? In analysis, many natural topologies are not metrizable. The space of bounded functions  $f : X \rightarrow \mathbb{R}$  with the topology of pointwise convergence ( $f_n \rightarrow f$  iff  $f_n(x) \rightarrow f(x)$  for all  $x$ ) does not come from a metric in general (it is a product topology). The  $C^\infty$  topology on smooth functions  $\mathbb{R} \rightarrow \mathbb{R}$  is also not metrizable. On the other hand, a metric contains extraneous information: the metrics  $d, d_1, d_\infty$  on  $\mathbb{R}^n$  all induce the same topology.

## 2.2 Continuous Maps Between Topological Spaces

**Definition 2.2.1: Continuous map**

A map  $f : X \rightarrow Y$  between topological spaces is continuous if  $f^{-1}(U)$  is open in  $X$  for every open set  $U \subseteq Y$ .

**Definition 2.2.2: Convergence in topological spaces**

A sequence  $\{p_n\}$  in a topological space  $X$  converges to  $p \in X$ , written  $p_n \rightarrow p$ , if for every open set  $U \ni p$ , there exists  $N$  such that  $p_n \in U$  for all  $n \geq N$ . Equivalently, every open set containing  $p$  contains all but finitely many  $p_n$ .

**Note:-**

In a general topological space, limits of sequences need not be unique. In the indiscrete topology  $\mathcal{T} = \{\emptyset, X\}$ , every sequence converges to every point (the only open set containing  $p$  is  $X$  itself). We will need the Hausdorff condition to guarantee uniqueness.

**Proposition 2.2.1** Basic properties of continuous maps

- (a) If  $f : X \rightarrow Y$  is continuous and  $p_n \rightarrow p$  in  $X$ , then  $f(p_n) \rightarrow f(p)$  in  $Y$ .
- (b) If  $f : X \rightarrow Y$  and  $g : Y \rightarrow Z$  are continuous, then  $g \circ f : X \rightarrow Z$  is continuous.

**Proof:** (a) Let  $V \ni f(p)$  be open in  $Y$ . Then  $f^{-1}(V)$  is open in  $X$  and contains  $p$ , so it contains  $p_n$  for all but finitely many  $n$ . Thus  $f(p_n) \in V$  for all but finitely many  $n$ .

(b) If  $W \subseteq Z$  is open, then  $g^{-1}(W)$  is open in  $Y$ , so  $f^{-1}(g^{-1}(W)) = (g \circ f)^{-1}(W)$  is open in  $X$ .  $\square$

### 2.2.1 Comparing Topologies

**Definition 2.2.3: Finer and coarser**

Given two topologies  $\mathcal{T}$  and  $\mathcal{T}'$  on  $X$ , we say  $\mathcal{T}'$  is finer than  $\mathcal{T}$  (or  $\mathcal{T}$  is coarser than  $\mathcal{T}'$ ) if  $\mathcal{T} \subseteq \mathcal{T}'$ . If  $\mathcal{T} \subsetneq \mathcal{T}'$ , we say  $\mathcal{T}'$  is strictly finer. Two topologies are comparable if one contains the other.

**Note:-**

The discrete topology is the finest topology on  $X$  (every point is isolated), and the indiscrete topology is the coarsest. A finer topology has more open sets: it is easier for functions into  $X$  to be continuous (more

preimages are open), but harder for sequences to converge (more open sets to check). In the discrete topology, a sequence converges iff it is eventually constant. In the indiscrete topology, every sequence converges to every point.

## 2.3 Bases for a Topology

Keeping track of all open sets is cumbersome. In metric spaces, we started with open balls and generated all open sets as unions. The abstract analogue is a basis.

### Definition 2.3.1: Basis for a topology

A basis for a topology on  $X$  is a collection  $\mathcal{B} \subseteq \mathcal{P}(X)$  satisfying:

- (i) For every  $x \in X$ , there exists  $B \in \mathcal{B}$  with  $x \in B$ .
- (ii) If  $x \in B_1 \cap B_2$  for  $B_1, B_2 \in \mathcal{B}$ , then there exists  $B_3 \in \mathcal{B}$  with  $x \in B_3 \subseteq B_1 \cap B_2$ .

The topology  $\mathcal{T}$  generated by  $\mathcal{B}$  consists of all sets that are unions of elements of  $\mathcal{B}$ . Equivalently,  $U \in \mathcal{T}$  iff for every  $x \in U$ , there exists  $B \in \mathcal{B}$  with  $x \in B \subseteq U$ .

#### Note:-

Condition (i) ensures  $X \in \mathcal{T}$  (since  $X = \bigcup_{B \in \mathcal{B}} B$ ). Condition (ii) ensures finite intersections of open sets are open: if  $U_1, U_2 \in \mathcal{T}$  and  $x \in U_1 \cap U_2$ , pick  $B_1, B_2 \in \mathcal{B}$  with  $x \in B_i \subseteq U_i$ , then (ii) gives  $B_3 \in \mathcal{B}$  with  $x \in B_3 \subseteq B_1 \cap B_2 \subseteq U_1 \cap U_2$ , so  $U_1 \cap U_2 \in \mathcal{T}$ . Unlike bases in linear algebra, topological bases can contain redundant elements—a better analogy is with generating sets.

### Example 2.3.1 (Bases for familiar topologies)

- The open balls  $\{B_r(x) : x \in X, r > 0\}$  form a basis for the metric topology on  $X$ . Even the rational-radius balls  $\{B_{1/n}(x) : x \in X, n \geq 1\}$  suffice.
- The open intervals  $\{(a, b) : a < b, a, b \in \mathbb{R}\}$  form a basis for the usual (standard) topology on  $\mathbb{R}$ . Condition (ii):  $(a_1, b_1) \cap (a_2, b_2) = (\max(a_1, a_2), \min(b_1, b_2))$  is either empty or another open interval.

### Definition 2.3.2: The lower limit topology

$\mathbb{R}_\ell$  is  $\mathbb{R}$  with the topology generated by the basis of half-open intervals  $\{[a, b) : a < b\}$ . This is strictly finer than the standard topology: every open interval  $(a, b) = \bigcup_n [a + 1/n, b)$  is open in  $\mathbb{R}_\ell$ , but  $[0, 1)$  is open in  $\mathbb{R}_\ell$  and not in the standard topology.

### Definition 2.3.3: The K-topology

$\mathbb{R}_K$  is  $\mathbb{R}$  with the topology generated by the basis  $\{(a, b) : a < b\} \cup \{(a, b) \setminus K : a < b\}$  where  $K = \{1/n : n \in \mathbb{Z}_{>0}\}$ . This is strictly finer than the standard topology but is not comparable to  $\mathbb{R}_\ell$ .

### Proposition 2.3.1 Continuity via bases

Let  $\mathcal{B}_X$  and  $\mathcal{B}_Y$  be bases for topologies on  $X$  and  $Y$ . Then  $f : X \rightarrow Y$  is continuous iff for every  $B \in \mathcal{B}_Y$  and every  $x \in f^{-1}(B)$ , there exists  $B' \in \mathcal{B}_X$  with  $x \in B' \subseteq f^{-1}(B)$ . In metric spaces, this reduces to the familiar  $\varepsilon$ - $\delta$  condition (with  $B = B_\varepsilon(f(x))$  and  $B' = B_\delta(x)$ ).

## 2.4 Subspace, Product, and Quotient Topologies

### 2.4.1 The Subspace Topology

#### Definition 2.4.1: Subspace topology

Let  $(X, \mathcal{T})$  be a topological space and  $Y \subseteq X$ . The subspace topology on  $Y$  is  $\mathcal{T}_Y = \{U \cap Y : U \in \mathcal{T}\}$ . That is, a set is open in  $Y$  iff it is the intersection of  $Y$  with an open set of  $X$ .

#### Note:-

When stating “ $U$  is open,” it is important to specify in which space. The set  $(0, 1)$  is open in  $\mathbb{R}$ , open in  $[0, 2]$  (since  $(0, 1) = (-1, 1) \cap [0, 2]$ ), but not open in  $\mathbb{R}^2$ . The set  $Y$  is always open in itself (as  $Y = X \cap Y$ ). If the metric on  $X$  restricts to  $Y$ , the subspace topology agrees with the induced metric topology: if  $\mathcal{T}_X$  comes from a metric  $d$  on  $X$ , then  $\mathcal{T}_Y$  comes from  $d|_{Y \times Y}$ .

#### Note:-

The subspace topology is the coarsest topology on  $Y$  making the inclusion  $\iota : Y \hookrightarrow X$  continuous.

### 2.4.2 The Product Topology

#### Definition 2.4.2: Product topology (finite case)

Given topological spaces  $(X, \mathcal{T}_X)$  and  $(Y, \mathcal{T}_Y)$ , the product topology on  $X \times Y$  is the topology generated by the basis

$$\mathcal{B} = \{U \times V : U \in \mathcal{T}_X, V \in \mathcal{T}_Y\}.$$

This is a basis: condition (i) holds since  $X \times Y \in \mathcal{B}$ , and condition (ii) since  $(U_1 \times V_1) \cap (U_2 \times V_2) = (U_1 \cap U_2) \times (V_1 \cap V_2) \in \mathcal{B}$ .

#### Note:-

When  $X, Y$  are metric spaces, the product topology on  $X \times Y$  is the metric topology for  $d_\infty((x_1, y_1), (x_2, y_2)) = \max(d_X(x_1, x_2), d_Y(y_1, y_2))$ . This agrees with the product topologies for the metrics  $d_2 = \sqrt{d_X^2 + d_Y^2}$  and  $d_1 = d_X + d_Y$ , since all three induce the same topology. The key observation is that  $B_r^{d_\infty}((x, y)) = B_r^X(x) \times B_r^Y(y)$ , so the metric balls are exactly the basis elements.

#### Proposition 2.4.1 Continuity into products

A map  $f : Z \rightarrow X \times Y$  is continuous iff both components  $f_1 = \pi_X \circ f : Z \rightarrow X$  and  $f_2 = \pi_Y \circ f : Z \rightarrow Y$  are continuous, where  $\pi_X, \pi_Y$  are the projection maps. Equivalently, a sequence  $(x_n, y_n) \rightarrow (x, y)$  in  $X \times Y$  iff  $x_n \rightarrow x$  in  $X$  and  $y_n \rightarrow y$  in  $Y$ .

**Proof:** The projections  $\pi_X, \pi_Y$  are continuous (preimage of  $U \subseteq X$  open is  $U \times Y$ , which is open). So if  $f$  is continuous, then  $f_1 = \pi_X \circ f$  and  $f_2 = \pi_Y \circ f$  are compositions of continuous maps. Conversely, if  $f_1, f_2$  are continuous and  $U \times V$  is a basis element of  $X \times Y$ , then  $f^{-1}(U \times V) = f_1^{-1}(U) \cap f_2^{-1}(V)$ , a finite intersection of open sets, hence open. Since every open set in  $X \times Y$  is a union of such basis elements,  $f^{-1}$  of any open set is open.  $\square$

### 2.4.3 Infinite Products: Box vs. Product Topology

For a family of spaces  $\{(X_i, \mathcal{T}_i)\}_{i \in I}$ , there are two natural topologies on  $\prod_{i \in I} X_i$ .

### Definition 2.4.3: Box topology

The box topology on  $\prod_{i \in I} X_i$  has basis consisting of all products  $\prod_{i \in I} U_i$  where each  $U_i \subseteq X_i$  is open.

### Definition 2.4.4: Product topology

The product topology on  $\prod_{i \in I} X_i$  has basis consisting of all products  $\prod_{i \in I} U_i$  where each  $U_i \subseteq X_i$  is open and  $U_i = X_i$  for all but finitely many  $i$ .

#### Note:-

For finite  $I$ , the two topologies coincide. For infinite  $I$ , the product topology is strictly coarser than the box topology. The product topology is the “right” choice: it is the coarsest topology making all projections  $\pi_i : \prod X_j \rightarrow X_i$  continuous. The characterization of continuity extends:  $f : Z \rightarrow \prod X_i$  is continuous (in the product topology) iff each component  $f_i = \pi_i \circ f$  is continuous.

### Example 2.4.1 (The diagonal map)

Let  $\Delta : \mathbb{R} \rightarrow \mathbb{R}^{\mathbb{N}}$  be defined by  $\Delta(x) = (x, x, x, \dots)$ . In the product topology,  $\Delta$  is continuous (each component  $\pi_n \circ \Delta = \text{id}_{\mathbb{R}}$  is continuous). In the box topology,  $\Delta$  is not continuous: the set  $U = \prod_{n=1}^{\infty} (-1/n, 1/n)$  is open in the box topology, but  $\Delta^{-1}(U) = \bigcap_{n=1}^{\infty} (-1/n, 1/n) = \{0\}$ , which is not open in  $\mathbb{R}$ .

## 2.4.4 Homeomorphisms

### Definition 2.4.5: Homeomorphism

Two topological spaces  $X$  and  $Y$  are homeomorphic, written  $X \cong Y$ , if there exist continuous maps  $f : X \rightarrow Y$  and  $g : Y \rightarrow X$  with  $g \circ f = \text{id}_X$  and  $f \circ g = \text{id}_Y$ . Equivalently,  $f$  is a bijection such that both  $f$  and  $f^{-1}$  are continuous. Such an  $f$  is called a homeomorphism. A homeomorphism is an isomorphism of topological spaces: it induces a bijection  $\mathcal{T}_X \leftrightarrow \mathcal{T}_Y$ .

#### Note:-

A continuous bijection need not be a homeomorphism: the inverse may fail to be continuous. If  $\mathcal{T}'$  is strictly finer than  $\mathcal{T}$  on  $X$ , then  $\text{id} : (X, \mathcal{T}') \rightarrow (X, \mathcal{T})$  is a continuous bijection (since  $\mathcal{T} \subseteq \mathcal{T}'$ ), but  $\text{id}^{-1} : (X, \mathcal{T}) \rightarrow (X, \mathcal{T}')$  is not continuous (some  $U \in \mathcal{T}' \setminus \mathcal{T}$  fails to pull back to an open set).

### Example 2.4.2

$f : (-\pi/2, \pi/2) \rightarrow \mathbb{R}$  defined by  $f(x) = \tan(x)$  is a homeomorphism (both  $\tan$  and  $\arctan$  are continuous). So  $\mathbb{R}$  is homeomorphic to any bounded open interval. This shows boundedness is not a topological property.

## 2.5 Closed Sets, Closure, Interior, Boundary

All the definitions from Lecture 1 carry over verbatim to general topological spaces.

### Definition 2.5.1: Closed set

A subset  $A$  of a topological space  $X$  is closed if  $X \setminus A$  is open. By De Morgan:  $\emptyset$  and  $X$  are closed, arbitrary intersections of closed sets are closed, and finite unions of closed sets are closed.

**Note:-**

A set can be both open and closed (clopen):  $\emptyset$  and  $X$  always are, and in the discrete topology every set is clopen. A set can also be neither open nor closed:  $[0, 1]$  in the standard topology on  $\mathbb{R}$ , or  $\mathbb{Q} \subset \mathbb{R}$ .

**Definition 2.5.2: Closure, interior, boundary**

Let  $A \subseteq X$  be any subset.

- The closure of  $A$  is  $\bar{A} = \bigcap \{F \supseteq A : F \text{ closed}\}$ , the smallest closed set containing  $A$ . A point  $x \in \bar{A}$  iff every open set containing  $x$  meets  $A$ .
- The interior of  $A$  is  $\text{Int}(A) = \bigcup \{U \subseteq A : U \text{ open}\}$ , the largest open set contained in  $A$ . A point  $x \in \text{Int}(A)$  iff some open set containing  $x$  is contained in  $A$ .
- The boundary of  $A$  is  $\partial A = \bar{A} \setminus \text{Int}(A)$ . A point  $x \in \partial A$  iff every open set containing  $x$  meets both  $\bar{A}$  and  $X \setminus A$ .

**Note:-**

The complementary identities hold:  $\overline{X \setminus A} = X \setminus \text{Int}(A)$  and  $\text{Int}(X \setminus A) = X \setminus \bar{A}$ . Also:  $A$  is closed iff  $\bar{A} = A$ , and  $A$  is open iff  $\text{Int}(A) = A$ .

**Example 2.5.1**

In  $\mathbb{R}$  with the standard topology: if  $A = [0, 1)$ , then  $\bar{A} = [0, 1]$ ,  $\text{Int}(A) = (0, 1)$ , and  $\partial A = \{0, 1\}$ .

**Definition 2.5.3: Neighborhood**

An open set  $U$  containing a point  $p$  is called a neighborhood of  $p$ . Then:  $p \in \text{Int}(A)$  iff  $A$  contains a neighborhood of  $p$ , and  $p \in \bar{A}$  iff every neighborhood of  $p$  intersects  $A$ .

**Definition 2.5.4: Dense subset**

$A$  is dense in  $X$  if  $\bar{A} = X$ , i.e., every nonempty open set intersects  $A$ . For instance,  $\mathbb{Q}$  is dense in  $\mathbb{R}$  (standard topology).

## 2.6 Limit Points and the Hausdorff Condition

### 2.6.1 Limit Points

**Definition 2.6.1: Limit point**

A point  $x \in X$  is a limit point of  $A \subseteq X$  if every neighborhood  $U$  of  $x$  satisfies  $U \cap (A \setminus \{x\}) \neq \emptyset$ . The set of limit points of  $A$  is denoted  $A'$ . A point of  $A$  that is not a limit point is an isolated point of  $A$ .

**Proposition 2.6.1 Closure via limit points**

$\bar{A} = A \cup A'$ . In particular,  $A$  is closed iff  $A' \subseteq A$  (iff  $A$  contains all its limit points).

**Proof:** If  $x \in A$ , every neighborhood meets  $A$  (it contains  $x$ ). If  $x \in A'$ , every neighborhood meets  $A \setminus \{x\} \subseteq A$ . Either way  $x \in \bar{A}$ . Conversely, if  $x \notin A$  and  $x \notin A'$ , then some neighborhood  $U$  satisfies  $U \cap (A \setminus \{x\}) = \emptyset$ ; since  $x \notin A$ , this gives  $U \cap A = \emptyset$ , so  $x \notin \bar{A}$ .  $\square$

### Example 2.6.1

In  $\mathbb{R}_{\text{std}}$ : 1 is a limit point of both  $(0, 1)$  and  $[0, 1]$ . In the set  $S = \{1/n : n \geq 1\} \cup \{0\}$ , the point 0 is a limit point (every neighborhood of 0 contains  $1/n$  for large  $n$ ), but 1 is an isolated point (the ball  $B_{1/2}(1) \cap S = \{1\}$ ).

#### Note:-

In metric spaces,  $x \in \overline{A}$  iff there exists a sequence in  $A$  converging to  $x$ , and  $x$  is a limit point of  $A$  iff there exists a sequence in  $A \setminus \{x\}$  converging to  $x$ . In general topological spaces, sequences are insufficient: the correct tool is nets or filters. However, in spaces with a countable basis of neighborhoods at each point (“first-countable” spaces, which include all metric spaces), sequences suffice.

## 2.6.2 Hausdorff Spaces

In the indiscrete topology on  $\mathbb{R}$ , every sequence converges to every point, and in the finite complement topology on  $\mathbb{R}$ , a sequence of distinct points converges to every point of  $\mathbb{R}$ . To exclude such pathologies:

### Definition 2.6.2: Hausdorff space

A topological space  $X$  is Hausdorff (or  $T_2$ ) if for every pair of distinct points  $x_1 \neq x_2 \in X$ , there exist disjoint open sets  $U_1, U_2$  with  $x_1 \in U_1$  and  $x_2 \in U_2$ .

### Theorem 2.6.1 Hausdorff implies unique limits

If  $X$  is Hausdorff, then every sequence in  $X$  converges to at most one limit.

**Proof:** Suppose  $x_n \rightarrow x$  and  $y \neq x$ . Choose disjoint open sets  $U_x \ni x$  and  $U_y \ni y$ . Since  $x_n \rightarrow x$ , there exists  $N$  with  $x_n \in U_x$  for all  $n \geq N$ . But then  $x_n \notin U_y$  for  $n \geq N$  (since  $U_x \cap U_y = \emptyset$ ), so  $\{x_n\}$  does not converge to  $y$ .  $\square$

### Example 2.6.2 (Hausdorff and non-Hausdorff spaces)

- (a) Every metric space is Hausdorff: if  $x \neq y$ , set  $r = d(x, y)/2 > 0$  and take  $U = B_r(x)$ ,  $V = B_r(y)$ .
- (b) The finite complement topology on  $\mathbb{R}$  is not Hausdorff: any two nonempty open sets have finite complements, so their intersection is  $\mathbb{R}$  minus a finite set, hence nonempty. But it is  $T_1$  (singletons are closed, since  $\mathbb{R} \setminus \{x\}$  has finite complement  $\{x\}$ ).
- (c) The discrete topology is always Hausdorff ( $U_i = \{x_i\}$ ).
- (d) Subspaces of Hausdorff spaces are Hausdorff. Products of Hausdorff spaces are Hausdorff.

#### Note:-

There is a hierarchy of separation axioms:  $T_1$  (points are closed)  $\Leftarrow T_2$  (Hausdorff)  $\Leftarrow T_3$  (regular)  $\Leftarrow T_4$  (normal). The finite complement topology on  $\mathbb{R}$  is  $T_1$  but not  $T_2$ . We will primarily work with Hausdorff spaces. Every metric space is  $T_4$  (hence satisfies all separation axioms).

## 2.7 Connectedness

### Definition 2.7.1: Connected space

A topological space  $X$  is connected if it cannot be written as  $X = U \cup V$  where  $U, V$  are disjoint, nonempty, open sets. Such a decomposition is called a separation of  $X$ . A subset  $A \subseteq X$  is connected if it is connected in the subspace topology.

### Proposition 2.7.1 $[0, 1]$ is connected

The interval  $[0, 1] \subset \mathbb{R}$  (with the standard topology) is connected.

**Proof:** Suppose  $[0, 1] = U \cup V$  is a separation with  $0 \in U$ . Let  $a = \sup\{x \in [0, 1] : [0, x] \subseteq U\}$ . Since  $U$  is open and  $0 \in U$ , we have  $a > 0$ .

If  $a \in V$ : since  $V$  is open in  $[0, 1]$ , there exists  $\varepsilon > 0$  with  $(a - \varepsilon, a] \cap [0, 1] \subseteq V$ . But by definition of  $a$ , the interval  $[0, a - \varepsilon)$  is not entirely in  $U$ , so some point of  $[0, a - \varepsilon)$  is in  $V$ . But then there exist points of  $V$  below  $a - \varepsilon$ , and  $[0, x] \subseteq U$  fails before  $a - \varepsilon$ , contradicting  $a = \sup$ .

More carefully: if  $a < 1$  and  $a \in U$ , then  $U$  is open, so  $(a - \varepsilon, a + \varepsilon) \subseteq U$  for some  $\varepsilon$ , contradicting  $a = \sup$ . If  $a \in V$ , then  $(a - \varepsilon, a + \varepsilon) \cap [0, 1] \subseteq V$ , so  $[0, x] \subseteq U$  for  $x < a - \varepsilon$  only, giving  $\sup \leq a - \varepsilon < a$ , contradiction. So  $a = 1$ , meaning  $U = [0, 1]$  and  $V = \emptyset$ .  $\square$

### Theorem 2.7.1 Continuous image of a connected space is connected

If  $f : X \rightarrow Y$  is continuous and  $X$  is connected, then  $f(X) \subseteq Y$  is connected.

**Proof:** Suppose  $f(X) = U \cup V$  is a separation (in the subspace topology). Then  $f^{-1}(U)$  and  $f^{-1}(V)$  are disjoint, nonempty (since  $U, V \subseteq f(X)$ ), and open in  $X$  (since  $f$  is continuous and  $U, V$  are open in  $f(X)$ , which means  $U = U' \cap f(X)$  for some open  $U' \subseteq Y$ , so  $f^{-1}(U) = f^{-1}(U')$  is open). Their union is  $X$ , giving a separation of  $X$ . Contradiction.  $\square$

### Corollary 2.7.1 Intermediate Value Theorem

Let  $X$  be a connected topological space and  $f : X \rightarrow \mathbb{R}$  continuous. If  $a, b \in X$  and  $r$  lies between  $f(a)$  and  $f(b)$ , then  $r \in f(X)$ : there exists  $c \in X$  with  $f(c) = r$ .

**Proof:** The image  $f(X)$  is a connected subset of  $\mathbb{R}$ . If  $r \notin f(X)$ , then  $U = f(X) \cap (-\infty, r)$  and  $V = f(X) \cap (r, \infty)$  give a separation of  $f(X)$  (both nonempty since  $f(a)$  is on one side and  $f(b)$  on the other). Contradiction.  $\square$

### Theorem 2.7.2 Connected subsets of $\mathbb{R}$ are intervals

A subset  $A \subseteq \mathbb{R}$  is connected (in the standard topology) if and only if  $A$  is an interval (i.e.,  $x, z \in A$  and  $x < y < z$  implies  $y \in A$ ).

## 2.7.1 Path-Connectedness

### Definition 2.7.2: Path

A path in  $X$  from  $x$  to  $y$  is a continuous map  $\gamma : [0, 1] \rightarrow X$  with  $\gamma(0) = x$  and  $\gamma(1) = y$ . (Here  $[0, 1]$  has the subspace topology from  $\mathbb{R}$ .)



**Definition 2.7.3: Path-connected**

A space  $X$  is path-connected if for every pair  $x, y \in X$ , there exists a path from  $x$  to  $y$ . The relation  $x \sim y$  (“connected by a path”) is an equivalence relation; the equivalence classes are the path components of  $X$ .

**Theorem 2.7.3** Path-connected implies connected

If  $X$  is path-connected, then  $X$  is connected.

**Proof:** Suppose  $X = U \cup V$  is a separation. Pick  $x \in U, y \in V$ , and let  $\gamma : [0, 1] \rightarrow X$  be a path from  $x$  to  $y$ . Then  $\gamma^{-1}(U)$  and  $\gamma^{-1}(V)$  are disjoint open sets covering  $[0, 1]$ , with  $0 \in \gamma^{-1}(U)$  and  $1 \in \gamma^{-1}(V)$ , giving a separation of  $[0, 1]$ . This contradicts the connectedness of  $[0, 1]$ .  $\square$

**Note:-**

The converse is false in general: the topologist’s sine curve  $S = \{(x, \sin(1/x)) : x > 0\} \cup \{(0, y) : -1 \leq y \leq 1\}$  is connected but not path-connected. However, for open subsets of  $\mathbb{R}^n$ , connected and path-connected are equivalent.

**Theorem 2.7.4** Products of connected spaces are connected

If  $X_i$  is connected for each  $i \in I$ , then  $\prod_{i \in I} X_i$  (with the product topology) is connected.

## 2.8 Exercises

**Exercise 2.8.1** Topology exercises

- (1) Verify that the finite complement topology  $\mathcal{T}_f$  on an infinite set  $X$  is  $T_1$  but not Hausdorff.
- (2) Let  $X$  be a set with topologies  $\mathcal{T}$  and  $\mathcal{T}'$ , with  $\mathcal{T}'$  strictly finer than  $\mathcal{T}$ . Show that  $\text{id} : (X, \mathcal{T}') \rightarrow (X, \mathcal{T})$  is continuous but not a homeomorphism.
- (3) Show that if  $X$  is Hausdorff, then  $Y \subseteq X$  (with the subspace topology) is Hausdorff, and  $X \times Y$  (with the product topology) is Hausdorff whenever both factors are.
- (4) Prove that  $A \subseteq \mathbb{R}$  (standard topology) is connected iff  $A$  is an interval.
- (5) Let  $X$  be a topological space and  $\{A_i\}_{i \in I}$  a family of connected subspaces with  $\bigcap_{i \in I} A_i \neq \emptyset$ . Show that  $\bigcup_{i \in I} A_i$  is connected.
- (6) Show that the closure of a connected set is connected.
- (7) Prove that a finite topological space is metrizable if and only if the topology is discrete.

**Solution**

- (1)  $T_1$ : for any  $x \in X$ ,  $X \setminus \{x\}$  has complement  $\{x\}$ , which is finite, so  $X \setminus \{x\} \in \mathcal{T}_f$ . Thus singletons are closed. Not Hausdorff: if  $U, V$  are nonempty open sets, then  $X \setminus U$  and  $X \setminus V$  are finite, so  $X \setminus (U \cap V) = (X \setminus U) \cup (X \setminus V)$  is finite. Since  $X$  is infinite,  $U \cap V$  is infinite (in particular nonempty). So no two nonempty open sets are disjoint.
- (2) Continuity: if  $U \in \mathcal{T}$ , then  $U \in \mathcal{T}'$  (since  $\mathcal{T} \subseteq \mathcal{T}'$ ), so  $\text{id}^{-1}(U) = U$  is open in  $\mathcal{T}'$ . Not a homeomorphism: since  $\mathcal{T}' \supsetneq \mathcal{T}$ , there exists  $V \in \mathcal{T}' \setminus \mathcal{T}$ . Then  $(\text{id}^{-1})^{-1}(V) = V \notin \mathcal{T}$ , so  $\text{id}^{-1}$  is not continuous.

(3) Subspace: if  $y_1 \neq y_2 \in Y \subseteq X$ , take disjoint open  $U_1, U_2 \subseteq X$  separating them. Then  $U_1 \cap Y, U_2 \cap Y$  are disjoint open sets in  $Y$  separating  $y_1, y_2$ . Product: given  $(x_1, y_1) \neq (x_2, y_2)$  in  $X \times Y$ , WLOG  $x_1 \neq x_2$ . Choose disjoint open  $U_1, U_2 \subseteq X$  with  $x_i \in U_i$ . Then  $U_1 \times Y$  and  $U_2 \times Y$  are disjoint open sets in  $X \times Y$  separating the two points.

(4) ( $\Rightarrow$ ) If  $A$  is not an interval, there exist  $x < y < z$  with  $x, z \in A$  and  $y \notin A$ . Then  $A = (A \cap (-\infty, y)) \cup (A \cap (y, \infty))$  is a separation. ( $\Leftarrow$ ) If  $A$  is an interval and  $A = U \cup V$  is a separation with  $u \in U, v \in V$ , WLOG  $u < v$ . Let  $a = \sup(U \cap [u, v])$ . If  $a \in U$ :  $U$  open gives  $(a - \varepsilon, a + \varepsilon) \cap A \subseteq U$ , contradicting  $a = \sup$  (there are points of  $U$  beyond  $a$ ). If  $a \in V$ :  $V$  open gives  $(a - \varepsilon, a + \varepsilon) \cap A \subseteq V$ , so  $\sup(U \cap [u, v]) \leq a - \varepsilon < a$ , contradiction.

(5) Suppose  $Y = \bigcup A_i = U \cup V$  is a separation. Pick  $p \in \bigcap A_i$ . WLOG  $p \in U$ . For each  $i$ , the sets  $U \cap A_i$  and  $V \cap A_i$  are disjoint open sets in  $A_i$  covering  $A_i$ , with  $p \in U \cap A_i \neq \emptyset$ . Since  $A_i$  is connected,  $V \cap A_i = \emptyset$  for all  $i$ . Thus  $V \cap Y = \bigcup (V \cap A_i) = \emptyset$ , contradicting  $V \neq \emptyset$ .

(6) Let  $A$  be connected and suppose  $\bar{A} = U \cup V$  is a separation. Then  $A = (U \cap A) \cup (V \cap A)$  with  $U \cap A$  and  $V \cap A$  open in  $A$  and disjoint. Since  $A$  is connected, one is empty, say  $V \cap A = \emptyset$ , so  $A \subseteq U$ . But then  $\bar{A} \subseteq \bar{U}$ . Since  $U$  is open in  $\bar{A}$  and  $V$  is its complement in  $\bar{A}$ ,  $U$  is also closed in  $\bar{A}$ , giving  $\bar{U} \cap \bar{A} = U$ . So  $\bar{A} \subseteq U$ , meaning  $V = \bar{A} \setminus U = \emptyset$ , contradiction.

(7) ( $\Leftarrow$ ) The discrete topology on a finite set  $X = \{x_1, \dots, x_n\}$  is metrizable (use the discrete metric). ( $\Rightarrow$ ) Suppose  $X$  is finite and metrizable with metric  $d$ . For each  $x \in X$ , let  $r_x = \min\{d(x, y) : y \in X, y \neq x\} > 0$  (the minimum exists since  $X$  is finite). Then  $B_{r_x}(x) = \{x\}$ , so singletons are open. Since every subset is a union of singletons, every subset is open, and the topology is discrete.